Target-Aware and Scale-Adaptive Distributional Conformal Prediction for Stochastic Simulation Outputs

Jin Zhao Xi Chen Meimei Liu

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Abstract

Predictive inference for stochastic simulation must provide reliable coverage for future simulation outputs while remaining informative under heteroscedastic and non-Gaussian response distributions. Distributional conformal prediction (DCP) is appealing because it calibrates a score based on an estimated conditional cumulative distribution function (CDF), but the coverage statement is only meaningful after specifying the input distribution over which future simulation scenarios are evaluated. We formulate simulation-output prediction relative to a user-specified target input distribution q_X, equivalently the target joint law $P^q_{XY}(dx, dy) = q_X(dx)P_{Y|X=x}(dy)$. This notation unifies ordinary global-domain coverage, region-of-interest coverage, and fixed-target-set average coverage. For the matched target-calibration setting, split conformal prediction gives finite-sample marginal validity under P^q_{XY}. To obtain efficient intervals under heterogeneous simulation noise, we use conditional kernel mean embeddings (CKME) to estimate the full conditional output law through smoothed indicator functions and introduce a scale-adaptive bandwidth $h(x) = c\widehat{s}(x)$, where $\widehat{s}(x)$ estimates the local response scale from training replications. Under a heteroscedastic location-scale model, oracle adaptive smoothing $h(x) = cs(x)$ removes the x-dependent standardized smoothing distortion created by a fixed bandwidth and restores homogeneity of the smoothed DCP score. Empirically, oracle adaptive smoothing improves worst-bin coverage deviation and paired interval score across Gaussian, queueing, and heavy-tailed benchmarks, while a robust plug-in estimator based on IQR smoothing closely tracks the oracle and preserves nominal marginal coverage.

1 Introduction

Stochastic simulation is routinely used to evaluate complex systems whose outputs are noisy, heteroscedastic, skewed, or heavy-tailed. In such settings, a predictive interval should answer a distributional question: for a future run at input x, what range of outputs Y should be expected with a prescribed probability? This question is not only about mean prediction. It also requires local information about the conditional response distribution $P_{Y|X=x}$, because simulation variability can change sharply across the input space.

Conformal prediction (CP) provides finite-sample, distribution-free coverage guarantees under exchangeability [19, 14, 7, 1]. Standard residual-based CP, however, calibrates a global residual scale. Under heteroscedasticity this can produce intervals that are too wide in low-variance regions and too narrow or unstable in high-variance regions. Distributional conformal prediction (DCP) addresses this issue by constructing conformity scores from an estimated conditional CDF [5]. If the conditional CDF is accurate, the probability integral transform maps responses onto a common probability scale, making interval construction more adaptive to local distributional shape.

For simulation output analysis, two issues remain underdeveloped. First, the coverage statement must specify the *target input distribution.* Simulation users often do not care about a natural population distribution over inputs; they may care about a design distribution over the whole experimental domain, a high-risk or policy-relevant region of interest, or a finite set of planned target scenarios. We denote this input target by q_X. The corresponding joint law is

$$P^q_{XY}(dx, dy) = q_X(dx)P_{Y|X=x}(dy),$$

and the intended coverage statement is marginal under this target joint law. Split conformal validity applies cleanly only when calibration inputs and target inputs are exchangeable under the same q_X. If calibration points are drawn from one design distribution and target points from another, the usual guarantee no longer follows without additional machinery such as weighted conformal calibration.

Second, when CKME is used to estimate a conditional CDF, the CDF is obtained by evaluating smoothed indicator functions. The smoothing bandwidth is not a harmless numerical detail. With a fixed bandwidth h, the standardized amount of smoothing is $h/s(x)$, where $s(x)$ is the local response scale. Thus the same scalar h oversmooths low-scale regions and undersmooths high-scale regions. This creates an avoidable distortion in the estimated DCP score, even if the conformal calibration step still protects marginal coverage.

This paper reframes CKME-DCP around these two principles: validity is target-distribution dependent, and efficiency is score-scale dependent. We use CKME as the conditional distribution estimator because it estimates a full conditional law as an RKHS-valued object and evaluates conditional CDFs through smooth test functions. This representation exposes a natural scale-adaptive smoothing rule,

$$h(x) = c\widehat{s}(x),$$

where $\widehat{s}(x)$ is estimated from training-stage simulation replications. The result is a conformal procedure that keeps the finite-sample split-CP guarantee at the target-distribution level while improving conditional efficiency through better local CDF smoothing.

The contributions are as follows.

1. We formulate CKME-DCP under a target joint law $P^q_{XY}(dx, dy) = q_X(dx)P_{Y|X=x}(dy)$. This makes clear whether coverage is claimed globally, over a region of interest, or on average over a fixed target set.

2. We introduce scale-adaptive CKME-DCP. It replaces a single smooth-indicator bandwidth with $h(x) = c\widehat{s}(x)$, where $\widehat{s}(x)$ is learned from training replications and smoothed across input space.

3. We separate validity from efficiency. Split conformal validity is estimator-agnostic and follows when calibration and target samples are exchangeable under q_X. Uniform CDF error transfers to DCP score and threshold error, affecting interval efficiency and group-conditional behavior.

4. We show that, under a heteroscedastic location-scale model, oracle adaptive smoothing $h(x) = cs(x)$ removes the x-dependent standardized smoothing ratio $h/s(x)$ induced by fixed smoothing and restores homogeneity of the smoothed-oracle DCP score.

5. We report numerical evidence from the current simulation suite. Oracle adaptive smoothing improves worst-bin coverage deviation and paired interval score on all considered data-generating processes. A robust IQR-based plug-in estimator closely tracks the oracle even under Student-t_3 noise.

2 Related Work

2.1 Conformal prediction and distributional scores

Conformal prediction constructs finite-sample prediction sets by calibrating a user-chosen nonconformity score on exchangeable data [19, 14]. Split conformal prediction separates model fitting from calibration and therefore applies to arbitrary predictive models, including black-box regression, Gaussian-process metamodels, distribution regression, and kernel-based conditional distribution estimators [7, 1]. This estimator-agnostic validity is central for simulation output analysis because the stochastic simulator may be expensive, nonlinear, and misspecified relative to any parametric metamodel.

The choice of score determines the geometry and efficiency of the resulting prediction set. Residual scores are simple and robust, but they often behave like a global-scale correction. Conformalized quantile regression uses estimated conditional quantiles to construct locally adaptive intervals [12]. Distributional conformal prediction instead estimates a conditional CDF and calibrates a score on the probability scale [5]. The present paper follows the distributional route because simulation analysts often need more than two quantiles: they may want the full conditional output distribution, interval inversion at multiple confidence levels, tail probabilities, or service-level probabilities. CKME is useful in this setting because it represents the conditional law as a single object rather than as a separate regression for every threshold or quantile.

2.2 Target, local, and weighted conformal inference

Conformal validity is a marginal statement over the distribution that generates the calibration and future examples. Recent work on weighted conformal prediction, covariate shift, and localized conformal prediction makes this dependence explicit by modifying calibration weights or neighborhoods when the target covariate distribution differs from the calibration distribution [16, 6]. Our target notation q_X is aligned with this literature, but our main operating regime is deliberately cleaner: the calibration and evaluation inputs are sampled from the same target distribution. This lets the paper keep the finite-sample theorem line simple while still making the scope of the guarantee visible.

This distinction is important in simulation. In observational prediction, the test covariate distribution may be viewed as given by nature. In simulation, the analyst usually chooses the design points, calibration points, and evaluation scenarios. A statement such as "90% coverage" is therefore incomplete unless it says whether the probability is over the whole design region, a safety-critical region of interest, a traffic peak, or a fixed portfolio of decision scenarios. Our contribution is not to solve every target-distribution mismatch problem. Rather, it is to put the target distribution into the formulation before introducing CKME-DCP and to separate matched-target validity from optional reweighting or local-calibration extensions.

2.3 Simulation metamodeling and uncertainty quantification

Classical simulation metamodeling focuses on learning the response surface of a stochastic simulator from designed experiments. Gaussian-process and stochastic-kriging methods are widely used for mean prediction and uncertainty quantification under simulation noise [13, 3]. Heteroscedastic Gaussian-process models further allow the noise variance to vary across the design space and are effective for many replication-based simulation settings [4]. These methods provide model-based predictive distributions or credible intervals, but their nominal uncertainty statements depend on the modeling assumptions.

Conformal calibration provides a complementary layer: it can wrap a fitted simulator metamodel or conditional distribution estimator and supply finite-sample marginal coverage under exchangeability. The price is that conformal calibration alone does not guarantee local efficiency. A poor score can still yield intervals that are too conservative in some regions and too aggressive in others. This paper therefore treats CKME and scale-adaptive smoothing as the score-construction layer, while split conformal calibration remains the validity layer.

2.4 Kernel embeddings for conditional distributions

Kernel mean embeddings represent probability distributions as elements of a reproducing kernel Hilbert space, and conditional kernel mean embeddings extend this representation to conditional laws [15, 8]. In the present application, the CKME representation is attractive because DCP needs repeated evaluations of $F(t \mid x)$ over a grid of candidate response values. Once the CKME weights are computed, the conditional expectation of a smooth test function can be evaluated by a weighted sum over training outputs.

The key technical issue is that the CDF indicator $1\{Y \leq t\}$ is typically not a smooth RKHS function. CKME-DCP therefore uses smooth indicator functions. This creates a bandwidth parameter on the output scale. Previous CKME-DCP use treats this bandwidth as a scalar tuning parameter. The present paper shows why that scalar choice is problematic under heteroscedasticity and proposes the scale-adaptive rule $h(x) = c\widehat{s}(x)$ as a direct correction to the standardized smoothing distortion.

3 Target-Aware Predictive Inference

3.1 Target input distributions and target joint laws

Let $X \in \mathcal{X} \subseteq \mathbb{R}^d$ denote a simulation input and $Y \in \mathbb{R}$ the corresponding stochastic output. We write $P_{Y|X=x}$ for the conditional output law at input x. Unlike many supervised-learning problems, the input distribution in simulation is often chosen by the analyst. We therefore distinguish three input distributions:

$$p_{\text{tr}}(x), \qquad p_{\text{cal}}(x), \qquad q_X(x),$$

where p_{tr} is the training design distribution used to fit the conditional CDF estimator, p_{cal} is the calibration design distribution, and q_X is the target distribution over which coverage is claimed.

Definition 1 (Target joint law and target coverage)**.** *For a user-specified target input distribution q_X, define*

$$P^q_{XY}(dx, dy) = q_X(dx)P_{Y|X=x}(dy).$$

A prediction set $\widehat{C}$ has marginal target coverage $1-\alpha$ if

$$\mathbb{P}_{(X,Y)\sim P^q_{XY}}\{Y \in \widehat{C}(X)\} \geq 1-\alpha.$$

This is not a new type of conformal validity; it is the standard marginal conformal statement evaluated under a user-specified target joint law. Different choices of q_X recover different simulation goals:

- *Global-domain target*: $q_X = P_X$, or a user-chosen space-filling distribution over the simulation domain, gives ordinary marginal coverage over that domain.

- *Region-of-interest target*: $q_X = P_X(\cdot \mid X \in \mathcal{R})$, or another distribution concentrated on $\mathcal{R}$, gives
$$\mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{R}\} \geq 1 - \alpha.$$
This is event-conditional or group coverage for the region $\mathcal{R}$, not pointwise conditional coverage at every $x \in \mathcal{R}$.

- *Fixed-target-set average*: for planned simulation scenarios $x_1^\star, \ldots, x_m^\star$, the discrete target
$$q_X = m^{-1} \sum_{j=1}^{m} \delta_{x_j^\star}$$
corresponds to the average scenario guarantee
$$\frac{1}{m} \sum_{j=1}^{m} \mathbb{P}_{Y \mid X = x_j^\star}\{Y \in \widehat{C}(x_j^\star)\} \geq 1 - \alpha.$$
This does not guarantee coverage at each individual scenario.

- *Smoothed fixed-target neighborhoods*: one may also define
$$q_\tau(x) = m^{-1} \sum_{j=1}^{m} K_\tau(x - x_j^\star)$$
and report a local guarantee around the target set rather than exactly on the discrete set.

This target notation prevents a common ambiguity. If calibration inputs are sampled globally but evaluation is restricted to a region of interest, the calibration and target pairs are generally not exchangeable. The usual split-conformal guarantee does not automatically transfer to that new target.

3.2 Matched target calibration as a design requirement

The notation $p_{\text{tr}}, p_{\text{cal}}, q_X$ also clarifies the roles of the different data sources in a simulation study. The training design p_{tr} controls where simulation budget is spent to learn the conditional distribution. It may be space-filling, adaptive, or concentrated where the simulator is difficult. The calibration distribution p_{cal}, however, determines the distribution of conformal scores used to set the prediction threshold. The evaluation or deployment target is q_X. For the standard split-conformal proof to apply without weighting, the calibration pairs and the future target pair must be exchangeable after conditioning on the fitted estimator.

This means that p_{tr} can differ from q_X without invalidating split conformal calibration, provided that the fitted estimator is treated as fixed during calibration and the calibration sample itself follows q_X. Such a separation is natural in simulation. One may spend the training budget broadly to learn the response surface, then spend the calibration budget on the operational region where guarantees are needed. Conversely, if calibration is global but evaluation is restricted to a high-risk region, the empirical coverage estimate is a diagnostic for that region, not a finite-sample conformal guarantee for that region.

The matched target-calibration regime is the main regime of this paper. It leads to the clean theorem line:
$$(X_i^{\text{cal}}, Y_i^{\text{cal}}),\, (X_\star, Y_\star) \overset{\text{exch.}}{\sim} P_{XY}^q \quad \Longrightarrow \quad \mathbb{P}\{Y_\star \in \widehat{C}(X_\star)\} \geq 1 - \alpha.$$

The statement is intentionally marginal under q_X. It is not a pointwise conditional guarantee, and it does not claim validity for target distributions that were not used in calibration. This boundary is important for the EV charging case study in Section 6.7, where q_X is a high-utilization region rather than the whole station-design space.

Remark 1 (Why keep p_{tr} separate?)**.** *If p_{tr} is too far from q_X, the CKME estimator may be inaccurate in the target region, producing wide or unstable intervals after calibration. This is an efficiency and estimation problem, not a conformal validity problem, as long as the calibration and target pairs are exchangeable. In practice, the training design should cover the support of q_X well enough for conditional distribution estimation, while the calibration design should match q_X for validity.*

3.3 Split conformal prediction and DCP scores

Let $Z = (X, Y) \in \mathcal{X} \times \mathcal{Y}$ denote an input-output pair drawn from the target joint law P^q_{XY}. For a new target input $X_\star$, we seek a set-valued predictor $\widehat{C}$ such that $Y_\star \in \widehat{C}(X_\star)$ with probability at least $1 - \alpha$. Conformal prediction provides such marginal coverage under exchangeability: conditional on any fitted model or score construction, the calibration pairs and the target pair must have a joint distribution invariant to permutations.

The construction relies on a nonconformity score $S : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, where larger values indicate that a candidate pair (x, y) is less compatible with the fitted model. A common regression score is the absolute residual $S(x, y) = |y - \widehat{\mu}(x)|$, but the conformal argument does not depend on this particular choice. Split conformal prediction separates model fitting from calibration. A training sample $\mathcal{D}_{\mathrm{tr}}$ is used to fit the predictive model or distribution estimator, and an independent calibration sample $\mathcal{D}_{\mathrm{cal}}$ is used only to calibrate the score. For a fitted score function $S(x, y)$, the calibration scores are

$$S_i^{\mathrm{cal}} = S(X_i^{\mathrm{cal}}, Y_i^{\mathrm{cal}}), \qquad i = 1, \ldots, n_{\mathrm{cal}}.$$

Let $\widehat{q}_{1-\alpha}$ be the $\lceil (n_{\mathrm{cal}} + 1)(1 - \alpha) \rceil$-th order statistic of these scores. The split conformal prediction set is

$$\widehat{C}(x) = \{y : S(x, y) \le \widehat{q}_{1-\alpha}\}.$$

Conditional on the fitted score, exchangeability of the calibration pair and the target pair is enough to obtain finite-sample marginal coverage. The score can be residual-based, quantile-based, localized, or distributional; the validity argument is the same.

Standard residual-based CP can be inefficient under heteroscedasticity because a single calibrated residual threshold tends to impose a nearly global response-scale correction. Distributional conformal prediction addresses this by using a score based on an estimated conditional CDF. Let

$$F(t \mid x) = \mathbb{P}(Y \le t \mid X = x)$$

denote the conditional CDF. Assume for motivation that $F(\cdot \mid x)$ is continuous for almost every x. Given an estimator $\widehat{F}(\cdot \mid x)$ fitted on $\mathcal{D}_{\mathrm{tr}}$, DCP defines a distributional nonconformity score at a candidate point (x, y). A common choice, and the one used in this paper, is

$$S_{\widehat{F}}(x, y) = \left| \widehat{F}(y \mid x) - 1/2 \right|. \tag{1}$$

More generally one may use $A(\widehat{F}(y \mid x))$ for a suitable transformation A. The score in (1) is motivated by the probability integral transform: if the true continuous CDF is known, then $F(Y \mid X)$ is uniformly distributed on $[0, 1]$ and no longer carries the original response scale. Thus DCP measures nonconformity on a probability scale rather than directly on the response scale, allowing prediction sets to adapt to heteroscedasticity, skewness, and tail behavior through the estimated conditional distribution.

Applying split conformal calibration to the DCP score yields

$$\widehat{C}_{\mathrm{DCP}}(x) = \{y \in \mathcal{Y} : S_{\widehat{F}}(x, y) \leq \widehat{q}_{1-\alpha}\}.$$

The conditional CDF estimator $\widehat{F}$ can be obtained by distribution regression, quantile-based methods, or other flexible conditional distribution estimators. Its accuracy primarily affects sharpness, interval shape, and groupwise behavior, while split conformal calibration supplies marginal validity under the target law. This is why the present paper separates the target-distribution validity layer from the CKME estimation and scale-adaptive smoothing layer.

3.4 Split conformal validity under a target distribution

Let $\mathcal{D}_{\mathrm{tr}}$ be a training sample used to fit a conditional CDF estimator $\widehat{F}$. Conditional on $\mathcal{D}_{\mathrm{tr}}$, let

$$(X_i^{\mathrm{cal}}, Y_i^{\mathrm{cal}}), \quad i = 1, \ldots, n_{\mathrm{cal}},$$

be independent calibration pairs from P_{XY}^q, i.e., $X_i^{\mathrm{cal}} \sim q_X$ and $Y_i^{\mathrm{cal}} \sim P_{Y|X=X_i^{\mathrm{cal}}}$. For a candidate pair (x, y), use the DCP score

$$S_{\widehat{F}}(x, y) = \left|\widehat{F}(y \mid x) - 1/2\right|. \tag{2}$$

Let $\widehat{q}_{1-\alpha}$ be the $\lceil (n_{\mathrm{cal}} + 1)(1 - \alpha) \rceil$-th order statistic of the calibration scores. The split DCP prediction set is

$$\widehat{C}(x) = \{y \in \mathbb{R} : S_{\widehat{F}}(x, y) \leq \widehat{q}_{1-\alpha}\}. \tag{3}$$

Proposition 1 (Target-distribution split conformal validity)**.** *Suppose that the calibration pairs and the target pair are exchangeable conditional on the fitted estimator $\widehat{F}$. In particular, this holds when calibration pairs and the target pair are independently drawn from the same target joint law P_{XY}^q. Then the prediction set in (3) satisfies*

$$\mathbb{P}_{(X,Y)\sim P_{XY}^q}\{Y \in \widehat{C}(X)\} \geq 1 - \alpha.$$

This guarantee does not require $\widehat{F}$ to be correctly specified.

Proposition 1 states the validity line of the paper. CKME and adaptive smoothing are not needed for marginal conformal validity. They affect the quality, shape, and conditional efficiency of the resulting intervals.

Remark 2 (When calibration and target distributions differ)**.** *If calibration inputs follow p_{cal} but target inputs follow q_X, unweighted split conformal calibration is generally calibrated for p_{cal}, not for q_X. Under support overlap, weighted conformal calibration can be used with weights proportional to $q_X(x)/p_{\mathrm{cal}}(x)$ [16]. This paper treats the weighted version as part of the target-aware formulation, while the main experiments use matched calibration and target distributions.*

Remark 3 (Fixed target inputs)**.** *A deterministic list of fixed target inputs is not automatically an exchangeable test sample from a global calibration design. If calibration and evaluation are both organized around the discrete target $m^{-1} \sum_j \delta_{x_j^\star}$, the conformal claim is an average over the target* *scenarios, not a separate guarantee for every $x_j^\star$. If nearby calibration points are more natural than exact replication at the targets, a practical alternative is to smooth the target set into*

$$q_\tau(x) = m^{-1} \sum_{j=1}^{m} K_\tau(x - x_j^\star),$$

collect or reweight calibration data for q_τ, and report coverage for that local target distribution. This is a target-distribution issue, not a bandwidth issue in the CKME CDF estimator.

4 Scale-Adaptive CKME-DCP

4.1 CKME conditional CDF estimation

Conditional kernel mean embedding provides a nonparametric representation of the full conditional output law. Rather than fitting a separate model for each threshold t, CKME embeds $P_{Y|X=x}$ as a single RKHS-valued object and evaluates conditional expectations through inner products [15, 8]. This is useful for DCP because the score requires the conditional CDF over many candidate response values when inverting the prediction set.

Let $\mathcal{H}_X$ and $\mathcal{H}_Y$ be RKHSs with kernels k_X on $\mathcal{X}$ and k_Y on $\mathcal{Y}$. We assume these kernels are bounded, e.g.,

$$\sup_{x\in\mathcal{X}} k_X(x,x) \le \kappa_X^2, \qquad \sup_{y\in\mathcal{Y}} k_Y(y,y) \le \kappa_Y^2.$$

Let $\phi_Y : \mathcal{Y} \to \mathcal{H}_Y$ denote the output feature map. Analogous to a kernel mean embedding for a marginal distribution, CKME represents the conditional distribution $P_{Y|X=x}$ by the conditional mean embedding

$$\mu_{Y|x} = \mathbb{E}[\phi_Y(Y) \mid X = x] \in \mathcal{H}_Y.$$

This representation is useful because, for any $g \in \mathcal{H}_Y$, the conditional expectation satisfies

$$\mathbb{E}[g(Y) \mid X = x] = \left\langle \mu_{Y|x}, g \right\rangle_{\mathcal{H}_Y}.$$

Hence CKME provides a compact distributional representation from which many conditional expectations can be recovered.

For conditional CDF estimation, $F(t \mid x) = \mathbb{P}(Y \le t \mid X = x)$ would correspond to $g(y) = \mathbf{1}\{y \le t\}$. Ideally one would write $F(t \mid x) = \left\langle \mu_{Y|x}, \mathbf{1}\{\cdot \le t\} \right\rangle_{\mathcal{H}_Y}$. However, this identity requires the indicator function to belong to $\mathcal{H}_Y$, which generally fails for smooth output kernels such as Gaussian or Matérn kernels. We therefore replace the hard indicator with a smooth surrogate.

Let $\Psi : \mathbb{R} \to [0,1]$ be nondecreasing and Lipschitz, approximating $\mathbf{1}\{u \ge 0\}$. For bandwidth $h > 0$, define

$$g_{t,h}(y) = \Psi\left(\frac{t-y}{h}\right).$$

The corresponding smoothed conditional CDF is

$$F_h(t \mid x) = \mathbb{E}[g_{t,h}(Y) \mid X = x] = \left\langle \mu_{Y|x}, g_{t,h} \right\rangle_{\mathcal{H}_Y},$$

provided $g_{t,h} \in \mathcal{H}_Y$. The bandwidth h therefore controls output-side smoothing in the CDF estimator.

Given training data $\mathcal{D}_{\mathrm{tr}} = \{(X_i, Y_i)\}_{i=1}^{n_{\mathrm{tr}}}$, CKME can be viewed as vector-valued kernel ridge regression for the conditional embedding. The fitted embedding has the closed form

$$\widehat{\mu}_{Y|x} = \sum_{i=1}^{n_{\mathrm{tr}}} a_i(x)\phi_Y(Y_i),$$

with weights

$$a(x) = (K_X + n_{\mathrm{tr}}\lambda I)^{-1} k_X(x),$$

where K_X is the $n_{\mathrm{tr}} \times n_{\mathrm{tr}}$ input Gram matrix with entries $[K_X]_{ij} = k_X(X_i, X_j)$, and

$$k_X(x) = \big(k_X(X_1, x), \dots, k_X(X_{n_{\mathrm{tr}}}, x)\big)^{\top}.$$

Here $\lambda > 0$ is the regularization parameter and I is the identity matrix. Substituting $\widehat{\mu}_{Y|x}$ into the smoothed CDF representation gives

$$\widehat{F}_h(t \mid x) = \langle \widehat{\mu}_{Y|x}, g_{t,h} \rangle_{\mathcal{H}_Y} = \sum_{i=1}^{n_{\mathrm{tr}}} a_i(x)\, \Psi\left(\frac{t - Y_i}{h}\right). \tag{4}$$

With an input-dependent bandwidth, we evaluate

$$\widehat{F}_{h(\cdot)}(t \mid x) = \sum_{i=1}^{n_{\mathrm{tr}}} a_i(x)\, \Psi\left(\frac{t - Y_i}{h(x)}\right). \tag{5}$$

The CKME weights $a(x)$ depend on the input kernel and regularization, not on the smoothing bandwidth. Thus $h(x)$ can be changed at prediction and calibration time without retraining the CKME coefficient system.

Remark 4 (Why not use hard empirical indicators?). *One can estimate $F(t \mid x)$ by replacing $g_{t,h}(Y_i)$ in (4) with $\mathbf{1}\{Y_i \leq t\}$. For each fixed t, this is a scalar kernel ridge regression problem for the binary response $\mathbf{1}\{Y \leq t\}$. It avoids smoothing bias, but it is no longer the CKME inner-product estimator in $\mathcal{H}_Y$, and it removes the bandwidth through which local response scale is controlled. Since the contribution of this paper is scale-adaptive score construction through $h(x)$, we use smooth indicators as the main estimator. Hard indicators are best treated as an ablation for calibrated interval performance, not as the main empirical race in conditional CDF estimation.*

4.2 Local scale and adaptive bandwidth

Here *scale* means the local spread of the simulation output distribution $Y \mid X = x$. In a location-scale model this is

$$Y = m(X) + s(X)\varepsilon, \qquad \varepsilon \perp X.$$

For Gaussian outputs, $s(x)$ is the conditional standard deviation; for Student-t outputs, it is the scale parameter multiplying the standardized noise. The fixed-bandwidth estimator uses the same response-scale smoothing everywhere. In standardized units, however, the effective bandwidth is

$$h_{\mathrm{eff}}(x) = \frac{h}{s(x)}.$$

Thus a fixed h can oversmooth low-noise regions, where $h/s(x)$ is large, and undersmooth high-noise regions, where $h/s(x)$ is small. This motivates the adaptive rule

$$h(x) = c\,\widehat{s}(x), \tag{6}$$

where $c > 0$ is a global multiplier and $\widehat{s}(x)$ is estimated from training-stage simulation replications. The goal is to make $h(x)/s(x)$ approximately stable across inputs.

In the current implementation, the training data are collected at design sites x_i with r_0 replications per site. At each site we compute the robust scale estimate

$$\widehat{s}_i = \frac{\mathrm{IQR}(Y_{i1}, \ldots, Y_{ir_0})}{1.349},$$

then smooth $(x_i, \widehat{s}_i)$ over $\mathcal{X}$ using Nadaraya-Watson regression with a Gaussian kernel. The resulting $\widehat{s}(x)$ is floored to avoid degenerate bandwidths, and $h(x) = c\widehat{s}(x)$ is used both for calibration scores and for prediction-set inversion.

The IQR estimator is not the only possible scale estimator. A sample standard deviation is efficient under Gaussian noise, but it can be unstable when replications are heavy-tailed or when a small number of extreme simulation outputs occur. A median absolute deviation estimator is another robust option. We use IQR because it has a simple Gaussian calibration constant, is stable for the Student-t example in the experiments, and is easy to compute from site-level replications. Since split conformal calibration absorbs moderate multiplicative scale errors, the purpose of $\widehat{s}(x)$ is not to estimate $s(x)$ perfectly. Its purpose is to remove large systematic changes in the standardized smoothing ratio across the input space.

The multiplier c controls the global amount of smoothing after scale normalization. When c is too small, the smoothed indicator becomes nearly discontinuous and the CKME CDF estimate can be noisy. When c is too large, the CDF estimate is biased toward an overly diffuse distribution and the DCP score loses resolution. In the experiments, a small sweep over c is used to verify that $c = 1$ is a stable default. A more elaborate implementation could tune c by cross-validation on a distributional loss, but we keep the default simple because the main contribution is the input-dependent scale normalization rather than another scalar tuning routine.

4.3 Prediction-set construction

The full scale-adaptive CKME-DCP procedure is:

1. Collect training data $\mathcal{D}_{\mathrm{tr}}$ and fit CKME weights with tuned (ℓ_x, λ).

2. Estimate the local scale function $\widehat{s}(x)$ from training replications and set $h(x) = c\widehat{s}(x)$.

3. For each calibration pair $(X_i^{\mathrm{cal}}, Y_i^{\mathrm{cal}})$, compute

$$\widehat{S}_i = \left| \widehat{F}_{h(\cdot)}(Y_i^{\mathrm{cal}} \mid X_i^{\mathrm{cal}}) - 1/2 \right|.$$

4. Let $\widehat{q}_{1-\alpha}$ be the split-conformal quantile of $\widehat{S}_1, \ldots, \widehat{S}_{n_{\mathrm{cal}}}$.

5. For a target input x, output

$$\widehat{C}(x) = \left\{ y \in \mathcal{T} : \left| \widehat{F}_{h(\cdot)}(y \mid x) - 1/2 \right| \le \widehat{q}_{1-\alpha} \right\},$$

   where $\mathcal{T}$ is the numerical grid used for inversion.

4.4 Implementation details

The implementation follows the training-calibration split used in the WSC version of CKME-DCP, with one change: the output-side bandwidth is evaluated as a function of the prediction input. The CKME hyperparameters (ℓ_x, λ) are selected using the training data. The fixed-bandwidth baseline also tunes a scalar h. The adaptive version replaces this scalar with $h(x) = c\widehat{s}(x)$, using the same CKME input weights $a(x)$. Thus the computationally expensive matrix solve for $(K_X + n_{\mathrm{tr}}\lambda I)^{-1}$ is unchanged.

Prediction-set inversion is performed on a response grid $\mathcal{T}$. In one-dimensional output settings this is straightforward: evaluate the DCP score $S_{\widehat{F}}(x,t)$ for every $t \in \mathcal{T}$ and return the grid values whose scores are below the calibrated threshold. The resulting set may be an interval or a union of intervals when the estimated CDF is imperfect. In the current numerical experiments we summarize it by the smallest interval covering the accepted grid values, which is the quantity used for coverage, width, and interval score. This convention matches the simulation-output setting where the user usually wants a one-dimensional prediction interval.

For repeated simulation outputs, there are two equivalent data views. One may treat each replication as an input-output pair (x_i, Y_{ij}), which is convenient for CKME fitting, while also preserving the site structure to estimate $\widehat{s}_i$ from the replications at each x_i. The local-scale estimator therefore uses the replication structure, but the conformal calibration scores are computed at the individual output level. The calibration sample is independent of the training sample used to fit CKME and estimate $\widehat{s}$.

The method has three distinct tuning components. The input kernel length scale controls how information is shared across neighboring simulation inputs. The regularization parameter controls the stability of the CKME weights. The output bandwidth controls smooth-indicator resolution. The proposed adaptive rule changes only the third component. This is important for interpreting the experiments: improvement over fixed-bandwidth CKME-DCP is evidence that the response-scale smoothing correction matters, rather than evidence that the entire CKME estimator has been replaced.

5 Theory

5.1 Estimator error and conformal behavior

Let $F^\circ(t \mid x)$ denote the oracle CDF target. Depending on context, this can be the exact conditional CDF or the smoothed oracle CDF corresponding to a chosen $h(x)$. Define the oracle score

$$S^\circ(x,y) = |F^\circ(y \mid x) - 1/2| \,.$$

All theoretical comparisons below are made over a response range $\mathcal{T}$ used for score evaluation and prediction-set inversion. We assume $\mathbb{P}(Y \in \mathcal{T}) = 1$ under the target law, or equivalently that $\mathcal{T}$ is chosen large enough that the omitted tail probability is negligible and handled outside the theorem.

Assumption 1 (Uniform CDF error)**.** *There exists $\epsilon_n = o_{\mathbb{P}}(1)$ such that*

$$\sup_{x\in\mathcal{X},\ t\in\mathcal{T}} \left|\widehat{F}(t \mid x) - F^\circ(t \mid x)\right| \le \epsilon_n$$

with probability tending to one.

Assumption 2 (Regular oracle score distribution)**.** *The marginal distribution of $S^\circ(X,Y)$, with $X \sim q_X$, is continuous and has a density g° bounded away from zero and infinity in a neighborhood of its $(1-\alpha)$-quantile $q^\circ_{1-\alpha}$.*

Proposition 2 (CDF error transfers to DCP score and threshold error)**.** *Under Assumptions 1–2,*

$$\sup_{x,y} \left| S_{\widehat{F}}(x,y) - S^{\circ}(x,y) \right| \leq \epsilon_n$$

with probability tending to one, and the split-conformal threshold satisfies

$$\left| \widehat{q}_{1-\alpha} - q^{\circ}_{1-\alpha} \right| = O_{\mathbb{P}}\left(\epsilon_n + n_{\text{cal}}^{-1/2} \right).$$

This proposition clarifies the role of the conditional CDF estimator. Finite-sample marginal validity comes from conformal calibration. CDF accuracy controls how close the calibrated prediction set is to an oracle DCP set and therefore affects width, interval score, and group-conditional coverage.

5.2 Why scale-adaptive smoothing helps

Assume the heteroscedastic location-scale model

$$Y = m(X) + s(X)\varepsilon, \qquad \varepsilon \perp X, \qquad s(x) > 0. \tag{7}$$

For fixed h, the smoothed oracle CDF is

$$F^{\circ}_{h}(t \mid x) = \mathbb{E}\left[\Psi\left(\frac{t - Y}{h} \right) \middle| X = x \right].$$

Writing $z = (t - m(x))/s(x)$, we have

$$F^{\circ}_{h}(t \mid x) = \mathbb{E}\left[\Psi\left(\frac{z - \varepsilon}{h/s(x)} \right) \right]. \tag{8}$$

Thus the standardized smoothing ratio $h/s(x)$ varies with x. A fixed h does not represent the same CDF resolution in low- and high-variance regions.

Now set $h(x) = cs(x)$. The smoothed oracle CDF becomes

$$F^{\circ}_{c}(t \mid x) = H_c\left(\frac{t - m(x)}{s(x)} \right), \qquad H_c(z) = \mathbb{E}\left[\Psi\left(\frac{z - \varepsilon}{c} \right) \right]. \tag{9}$$

Theorem 1 (Smoothed-score homogeneity under oracle adaptive bandwidth)**.** *Suppose (7) holds and $h(x) = cs(x)$. For any measurable score transform $A : [0,1] \to \mathbb{R}$, define*

$$S^{\circ}_{c}(x,y) = A(F^{\circ}_{c}(y \mid x)).$$

Then

$$S^{\circ}_{c}(X,Y) \mid X = x \overset{d}{=} A(H_c(\varepsilon)).$$

Therefore the conditional distribution of the smoothed-oracle score does not depend on x.

Theorem 1 is the central scale-adaptive argument. Exact DCP with the true continuous CDF already has probability-integral-transform homogeneity. CKME-DCP uses smoothed indicators, and fixed smoothing can break this homogeneity through $h/s(x)$. Adaptive smoothing restores it at the smoothed-oracle level.

Assumption 3 (Uniform relative scale consistency)**.** *The plug-in scale estimator satisfies*

$$\Delta_n = \sup_{x\in\mathcal{X}} \left| \frac{\widehat{s}(x)}{s(x)} - 1 \right| = o_{\mathbb{P}}(1).$$

Assumption 4 (Standardized moment bound)**.** *For the response range $\mathcal{T}$,*

$$M_{\mathcal{T}} = \sup_{x\in\mathcal{X},\ t\in\mathcal{T}} \mathbb{E}\left[\left| \frac{t-m(x)}{s(x)} - \varepsilon \right| \middle| X = x \right] < \infty.$$

The smooth step function Ψ is Lipschitz with constant L_Ψ.

Proposition 3 (Plug-in adaptive bandwidth is close to oracle adaptive bandwidth)**.** *Suppose the location-scale model* (7) *holds and Assumptions 3–4 hold. Let*

$$F^\circ_{\widehat{h}}(t \mid x) = \mathbb{E}\left[\Psi\left(\frac{t-Y}{c\widehat{s}(x)} \right) \middle| X = x \right]$$

and let F°_c be the oracle adaptive smoothed CDF in (9)*. Then*

$$\sup_{x\in\mathcal{X},\ t\in\mathcal{T}} \left| F^\circ_{\widehat{h}}(t \mid x) - F^\circ_c(t \mid x) \right| = O_{\mathbb{P}}(\Delta_n) = o_{\mathbb{P}}(1).$$

5.3 CKME consistency and group-conditional behavior

Let the smoothed oracle target be

$$F^\circ_{h(\cdot)}(t \mid x) = \mathbb{E}[g_{t,h(x)}(Y) \mid X = x].$$

Assumption 5 (Uniform CKME embedding consistency)**.** *For the chosen kernels and regularization sequence,*

$$\sup_{x\in\mathcal{X}} \left\| \widehat{\mu}_{Y|x} - \mu_{Y|x} \right\|_{\mathcal{H}_Y} = o_{\mathbb{P}}(1).$$

Assumption 6 (Bounded smoothed-indicator class)**.** *For the target range $\mathcal{T}$,*

$$\sup_{x\in\mathcal{X},\ t\in\mathcal{T}} \left\| g_{t,h(x)} \right\|_{\mathcal{H}_Y} < \infty.$$

Assumption 7 (Finite-partition oracle score regularity)**.** *Let $\mathcal{G}_1, \ldots, \mathcal{G}_K$ be a fixed finite partition of $\mathcal{X}$, with $\mathbb{P}(X \in \mathcal{G}_k) > 0$. For the oracle adaptive score $S^\circ_c(X,Y)$, the groupwise distribution functions*

$$G_k(q) = \mathbb{P}\{S^\circ_c(X,Y) \le q \mid X \in \mathcal{G}_k\}$$

are Lipschitz in a common neighborhood of $q^\circ_{1-\alpha}$, uniformly over $k = 1, \ldots, K$.

Proposition 4 (CKME embedding consistency implies smoothed-CDF consistency)**.** *Under Assumptions 5–6,*

$$\sup_{x\in\mathcal{X},\ t\in\mathcal{T}} \left| \widehat{F}_{h(\cdot)}(t \mid x) - F^\circ_{h(\cdot)}(t \mid x) \right| = o_{\mathbb{P}}(1).$$

Combining Proposition 4 with Proposition 2 gives score and threshold stability for CKME-DCP. Combining those with Theorem 1 yields the following group-conditional implication.

Theorem 2 (Asymptotic finite-partition group coverage). *Suppose the location-scale model holds, oracle adaptive smoothing $h(x) = cs(x)$ is used, and Assumptions 1, 2, 5, 6, and 7 hold with $F^\circ = F^\circ_c$. Then*

$$\max_{1\le k\le K} \left|\mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{G}_k\} - (1-\alpha)\right| \to 0$$

in probability.

Corollary 1 (Plug-in scale-adaptive CKME-DCP). *Suppose the location-scale model holds, the plug-in bandwidth $h(x) = c\widehat{s}(x)$ is used, and Assumptions 2, 3, 4, 5, 6, and 7 hold. If the CKME estimator is uniformly consistent for the plug-in smoothed oracle $F^\circ_{\widehat{h}}$, then the conclusion of Theorem 2 continues to hold:*

$$\max_{1\le k\le K} \left|\mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{G}_k\} - (1-\alpha)\right| \to 0$$

in probability.

The theorem does not claim finite-sample pointwise conditional coverage, which is impossible in general without stronger assumptions. It states that, in the location-scale regime targeted here, adaptive smoothing plus uniform CKME CDF consistency can make DCP scores asymptotically homogeneous over fixed finite input partitions.

6 Numerical Evidence

6.1 Experimental setup

The current experiments use the training-calibration split inherited from the WSC version of the paper. Stage 1 trains CKME and estimates local scale from simulation replications. Stage 2 supplies calibration data for split DCP. In these experiments the calibration and evaluation inputs are matched to the same global-domain target q_X, so the empirical coverage is interpreted under the target joint law P^q_{XY}. The target marginal coverage is $1 - \alpha = 0.90$.

The adaptive-bandwidth experiments cover four one-dimensional data-generating processes:

`wsc_gauss` Smooth heteroscedastic Gaussian output with $s(x) = 0.01 + 0.20(x - \pi)^2$.

`gibbs_s1` Heteroscedastic Gaussian output with $s(x) = |\sin x|$.

`exp1` An M/G/1 queueing-inspired model with nonlinear scale behavior near the boundary.

`nongauss_A1L` A Student-t_3 heavy-tailed output with the same smooth scale function as `wsc_gauss`.

We report marginal coverage, mean width, Winkler interval score, and bin-wise coverage diagnostics. Conditional performance is summarized by the worst-bin deviation $\max_b |\widehat{c}_b - (1-\alpha)|$, where $\widehat{c}_b$ is empirical coverage in bin b. Macrorep-paired comparisons use common random numbers across fixed and adaptive bandwidth arms.

6.2 Benchmark structure and evaluation protocol

The numerical section is organized around the paper's main question rather than around an exhaustive leaderboard. The key ablation is fixed-bandwidth CKME-DCP versus scale-adaptive CKME-DCP. Both methods use the same training data, the same calibration data, the same CKME input kernel family, and the same conformal calibration rule. The difference is the output-side smooth-indicator bandwidth. This paired design isolates the effect of $h(x) = c\widehat{s}(x)$.

For a full journal version, the benchmark set should contain three layers:

Table 1: Benchmark roles for the full journal version. The present draft has numerical results for the internal CKME ablations; the remaining rows define the planned comparison structure.

Method class	Examples	Role in the paper
Internal CKME ablation	Fixed CKME-DCP, oracle adaptive CKME-DCP, plug-in adaptive CKME-DCP	Isolate the effect of scale-adaptive smooth-indicator bandwidths.
External conformal	DCP-DR, CQR, residual CP	Compare against conformal methods with different score constructions.
Simulation UQ	hetGP, stochastic kriging, conformalized GP intervals	Connect to the simulation metamodeling literature and distinguish model-based from conformal uncertainty.
Target-aware calibration	Matched ROI calibration, weighted calibration if used	Demonstrate that coverage is evaluated under the stated q_X.

Internal CKME ablations. Fixed-bandwidth CKME-DCP, oracle scale-adaptive CKME-DCP, and plug-in scale-adaptive CKME-DCP. These arms directly test the proposed mechanism.

Conformal competitors. DCP-DR and CQR are natural external conformal baselines. DCP-DR estimates the conditional distribution through threshold-wise distribution regression, while CQR estimates conditional quantiles. These methods are useful because they ask whether CKME-DCP is competitive with established conformal approaches that do not rely on CKME smooth indicators.

Simulation UQ baselines. Heteroscedastic GP or stochastic-kriging intervals are useful for the simulation audience. When reported without conformal calibration, they should be described as model-based uncertainty intervals rather than distribution-free conformal intervals. A conformalized version can also be included if implementation time permits.

The current draft includes the internal CKME ablations because they are the evidence needed for the scale-adaptive bandwidth claim. The external conformal and simulation-UQ baselines should be added once their implementations are stable enough to avoid distracting from the main contribution. In the EV case study, the same principle applies: the first comparison should be fixed versus scale-adaptive CKME-DCP under the same ROI target distribution. External baselines can then be used to show that the proposed method is not merely improving over a weak CKME variant.

Across all experiments, we report both marginal and conditional diagnostics. Marginal coverage is necessary because it is the formal conformal target. Mean width and Winkler interval score measure efficiency and penalize undercoverage. Bin-wise coverage and worst-bin deviation are diagnostic rather than guaranteed finite-sample quantities; they show whether a method distributes coverage errors evenly across the input domain or hides local failures behind a correct marginal average. For a top-journal version, these diagnostics should be reported with macroreplication uncertainty, such as standard errors or paired confidence intervals for method differences.

Table 2: Oracle adaptive bandwidth versus fixed bandwidth. The table is generated from the current adaptive-bandwidth experiment. Δ is oracle minus fixed, so negative worst-bin deviation is better.

	Worst-bin $\lvert c-(1-\alpha)\rvert$			Mean width			Marginal cov	
DGP	fixed	oracle	Δ	fixed	oracle	ratio	fixed	oracle
wsc_gauss	0.148	0.100	-0.048	2.126	2.114	0.99	0.900	0.902
gibbs_s1	0.106	0.098	-0.008	2.200	2.189	1.00	0.903	0.901
exp1	0.159	0.091	-0.068	2.099	2.400	1.14	0.898	0.898
nongauss_A1L	0.136	0.105	-0.032	3.179	2.988	0.94	0.903	0.901

Table 3: Macrorep-paired improvements from oracle adaptive smoothing. Negative values indicate oracle is better for deviation, width, or interval score.

DGP	Δ worst-bin	Δ width	Δ interval score	Oracle better IS
`wsc_gauss`	-0.033	-0.008	-0.125	1.00
`gibbs_s1`	-0.016	-0.008	-0.069	1.00
`exp1`	-0.061	0.300	-0.215	0.98
`nongauss_A1L`	-0.021	-0.164	-0.290	0.98

6.3 Oracle adaptive bandwidth

The oracle arm sets $h(x) = cs(x)$, with $c = 1$ unless otherwise stated. The fixed arm uses a CV-tuned scalar bandwidth. Table 2 shows that oracle adaptive smoothing reduces worst-bin coverage deviation on all four DGPs while preserving marginal coverage near 0.90. The paired interval-score difference is also consistently favorable: oracle adaptive smoothing wins the paired interval score in 98–100% of macroreplications across the four DGPs.

6.4 Choice of the multiplier c

The multiplier c controls the smoothing scale relative to the local response scale. A sweep on `nongauss_A1L` shows that worst-bin coverage deviation improves from the fixed baseline 0.136 to 0.105 at $c = 1$, with little additional gain at $c = 2$. We therefore use $c = 1$ as the default in the plug-in experiments.

6.5 Plug-in scale estimation

The oracle $s(x)$ is unavailable in practice. We estimate it from Stage-1 replications using IQR/1.349 at each design site and smooth these sitewise values over x. Table 5 summarizes the plug-in gap at the largest Stage-1 budget $B = 500$. The absolute coverage gap between plug-in and oracle is at the macrorep noise floor, between 0.006 and 0.008 across the four DGPs. The IQR plug-in has a predictable finite-r_0 shrinkage under Gaussian noise and remains stable under Student-t_3 noise.

6.6 Robust scale versus sample standard deviation

Table 6 compares sample standard deviation and IQR-based plug-in scale estimators at $B = 500$. Both yield similar downstream conformal coverage, because split conformal calibration absorbs

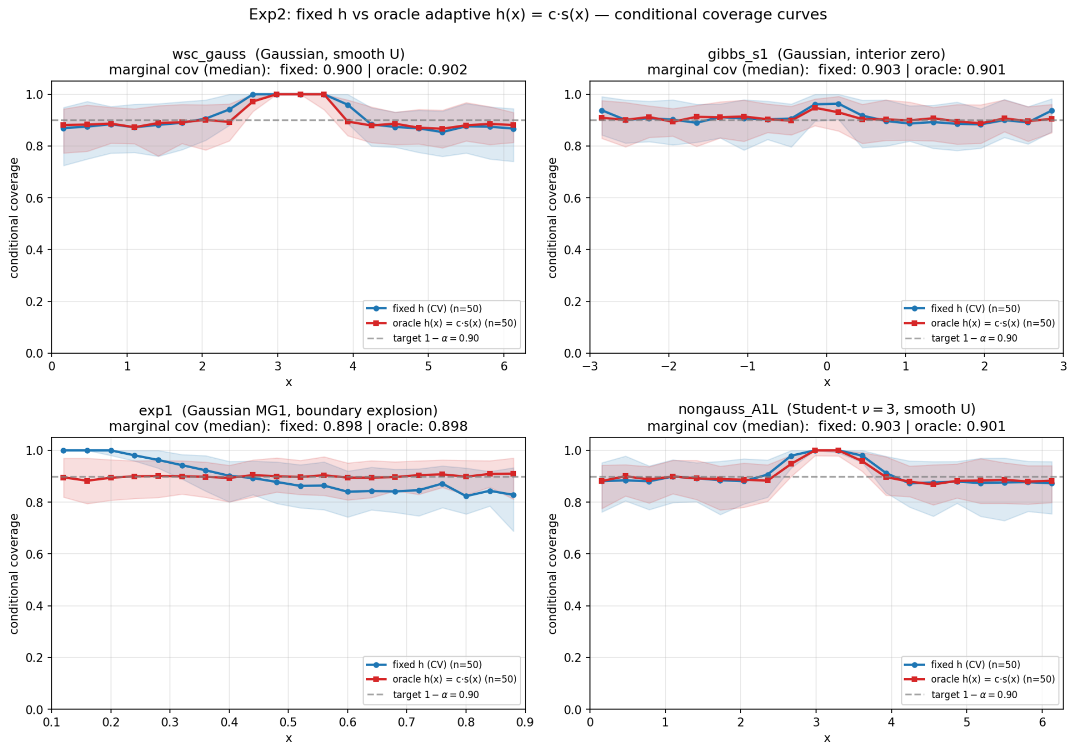

Figure 1: Bin-wise coverage curves for fixed and oracle adaptive smoothing. Oracle adaptive smoothing flattens coverage across input bins while keeping marginal coverage near the conformal target.

moderate multiplicative scale offsets. The difference is theoretical and diagnostic: the sample SD is unstable under Student-t_3 noise, producing a bandwidth ratio of 1.418, whereas IQR remains close to its finite-sample target with ratio 0.955. We therefore use the IQR estimator as the default robust scale estimator.

6.7 Planned case study: EV fast-charging congestion

To connect the method to a realistic stochastic simulation problem, we will add a case study on prediction intervals for waiting time at public EV fast-charging stations under peak corridor-demand scenarios. EV charging infrastructure is a timely application area because public charging demand, station sizing, capacity analysis, and grid impacts are explicitly studied through high-resolution road-trip charging simulation tools such as EVI-RoadTrip [10]. Public station reliability and availability are also linked to EV adoption concerns [11]. Recent queueing and planning studies further emphasize congestion, waiting-time service levels, and uncertain demand as central operational issues for EV charging networks [18, 17].

6.7.1 Operational question

The operational question is not merely whether the expected waiting time is large. A planner needs uncertainty-aware statements such as: under peak travel demand and a specified station configuration, what interval should contain the next customer's waiting time with 90% target coverage? This question is naturally distributional. Waiting time is often close to zero in low-

Table 4: c-sweep on `nongauss_A1L`. The fixed row is the scalar-bandwidth baseline.

arm	c	marginal cov	worst-bin $\lvert c-(1-\alpha)\rvert$	mean-bin $\lvert c-(1-\alpha)\rvert$	mean width	mean IS
fixed	—	0.903 (0.013)	0.136 (0.041)	0.056 (0.009)	3.179 (0.275)	5.549 (0.774)
c0.3	0.3	0.899 (0.013)	0.115 (0.034)	0.052 (0.008)	3.129 (0.226)	5.410 (0.758)
c0.5	0.5	0.901 (0.012)	0.113 (0.028)	0.050 (0.007)	3.037 (0.178)	5.341 (0.752)
c1	1	0.901 (0.012)	0.105 (0.021)	0.046 (0.007)	2.988 (0.150)	5.276 (0.744)
c2	2	0.900 (0.012)	0.104 (0.020)	0.042 (0.007)	3.009 (0.153)	5.251 (0.742)

Table 5: Plug-in versus oracle at Stage-1 budget $B = 500$. Coverage gap is the median absolute difference between plug-in and oracle marginal coverage over macroreplications.

DGP	Coverage gap	q-ratio	h-ratio	Width ratio
`exp1`	0.006	0.952	0.852	0.853
`gibbs_s1`	0.007	1.085	0.875	1.028
`nongauss_A1L`	0.006	0.962	0.955	0.924
`wsc_gauss`	0.008	1.055	0.836	0.937

utilization regimes, but it can become highly variable and right-skewed as utilization approaches one. A mean metamodel can miss this transition, and a fixed-scale conformal correction can hide local instability behind marginal coverage.

This case study therefore matches the paper's two themes. First, the relevant target distribution is not the full design region. It is an operational region of high congestion where reliability matters. Second, the conditional response scale changes sharply across that region, making fixed output-side smoothing a plausible source of DCP score distortion. The case study should demonstrate these two points in a setting that is recognizable to simulation and operations researchers.

The case study is designed to illustrate the target-aware part of the paper without changing the main theorem line. The target input distribution is not the entire design space. Instead, it concentrates on the region of high-utilization operating conditions where planners care most about service reliability:

$$q_X = p_X\{\,\cdot \mid \rho(X) \in [0.75, 0.98]\,\},$$

where $\rho(X)$ is the implied station utilization under input setting X. Calibration and evaluation inputs will be sampled from this same ROI target distribution q_X, so the empirical coverage claim remains the clean matched target-calibration claim of Proposition 1. We do not use this case study to claim validity for arbitrary deployment distributions.

6.7.2 Discrete-event simulation model

The simulator will be a discrete-event multi-server charging-station queue, such as an $M/G/c$ or $M/G/c/K$ model. Inputs may include an arrival-rate multiplier, number of available DC fast-charging plugs, mean energy demand per vehicle, charging power or mean service rate, plug availability/reliability, and a peak-travel indicator. Public station and port data from the Alternative Fuel Stations API can be used to ground realistic ranges for station size and charger availability [9]. The primary output Y is customer waiting time before charging. Secondary outputs, if needed, include total time to complete charging or a service-level violation indicator.

One concrete input vector is

$$X = (\eta, c, \bar{e}, p_{\text{chg}}, a, \tau),$$

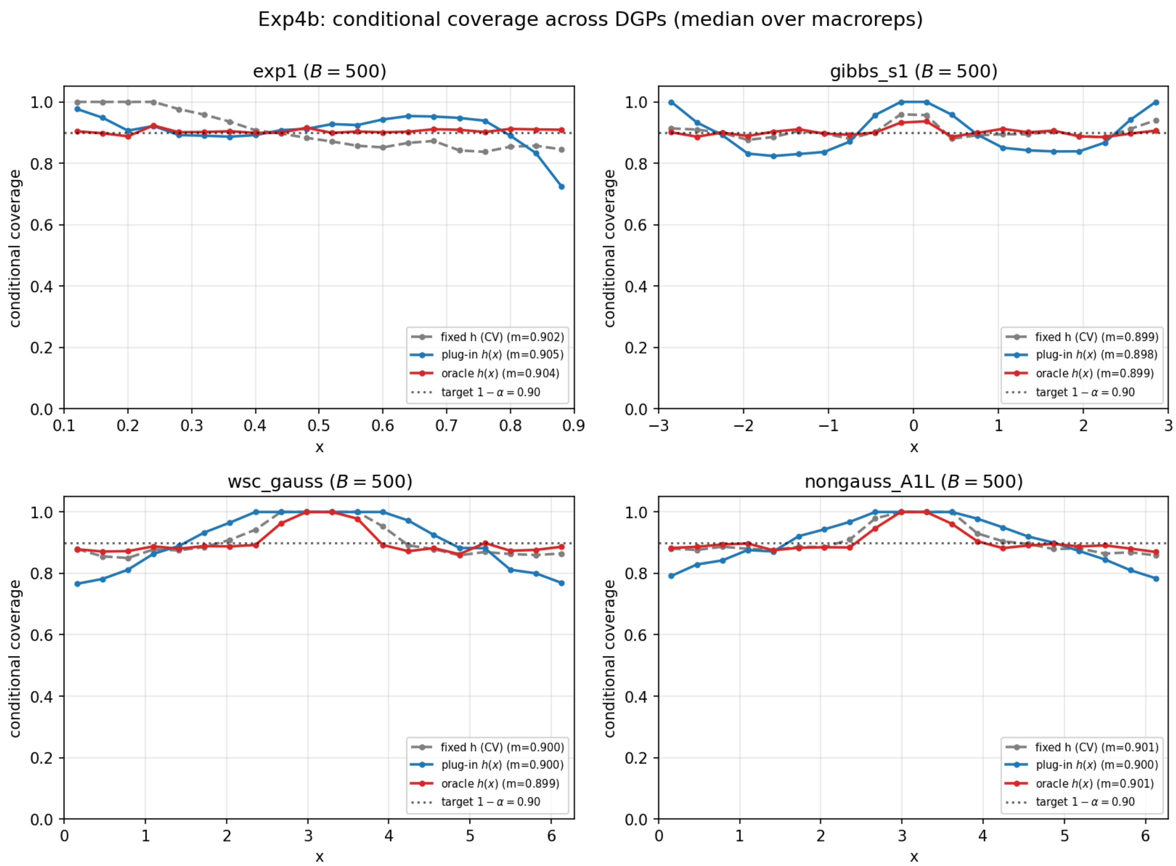

Figure 2: Coverage curves at $B = 500$ for fixed, oracle, and plug-in bandwidths. Plug-in and oracle adaptive smoothing nearly overlap across the four DGPs.

where η is an arrival-rate multiplier relative to a baseline corridor demand, c is the number of DC fast-charging plugs, $\bar{e}$ is the mean energy demand per arriving vehicle, p_{chg} is the charging power or effective service-rate parameter, a is the plug availability probability, and τ indicates a peak travel period. Given X, arrivals can be generated by a Poisson or nonhomogeneous Poisson process over a finite horizon, energy demands can be sampled from a positive right-skewed distribution, and plug failures can be sampled at the scenario or run level. The implied utilization $\rho(X)$ is computed from the arrival rate, effective service demand, number of available plugs, and reliability adjustment.

The case study should be implemented as a stochastic simulator rather than as a closed-form queueing formula. This is important because the paper is about simulation output prediction. The simulator can include finite horizon effects, random plug availability, and non-exponential charging times. These features make the conditional waiting-time law non-Gaussian and heteroscedastic, while still being simple enough for controlled macroreplication studies.

6.7.3 Experimental design

The training design should cover a broader feasible region than the target ROI so that CKME can learn how the waiting-time distribution changes across utilization levels. The calibration and evaluation designs should be sampled from the ROI target q_X. This creates a clean demonstration

Table 6: SD versus IQR plug-in scale estimator at $B = 500$.

	Sample SD				IQR/1.349			
DGP	$\bar{h}_{\mathrm{p}}/\bar{h}_{\mathrm{o}}$	$\lvert\Delta\mathrm{cov}\rvert$	$\mathrm{cov}_{\mathrm{plug}}$	$\mathrm{cov}_{\mathrm{or}}$	$\bar{h}_{\mathrm{p}}/\bar{h}_{\mathrm{o}}$	$\lvert\Delta\mathrm{cov}\rvert$	$\mathrm{cov}_{\mathrm{plug}}$	$\mathrm{cov}_{\mathrm{or}}$
exp1	0.961	0.0060	0.903	0.902	0.852	0.0060	0.904	0.902
gibbs_s1	0.972	0.0080	0.896	0.897	0.875	0.0070	0.896	0.897
wsc_gauss	0.926	0.0080	0.899	0.900	0.836	0.0080	0.899	0.900
nongauss_A1L	1.418	0.0080	0.900	0.902	0.955	0.0060	0.901	0.902

of the distinction between training design and target calibration:

$$p_{\mathrm{tr}} \neq q_X, \qquad p_{\mathrm{cal}} = q_X, \qquad p_{\mathrm{eval}} = q_X.$$

The paper can then state that target validity is claimed for the ROI because calibration and evaluation match under q_X, while the broader training design is used only to improve the conditional CDF estimator.

The design should include utilization bins inside the ROI, for example $[0.75, 0.82)$, $[0.82, 0.89)$, $[0.89, 0.94)$, and $[0.94, 0.98]$. These bins are not separate finite-sample coverage guarantees. They are diagnostic groups used to see whether a method has stable behavior across increasingly congested regimes. A useful figure would plot the estimated scale $\widehat{s}(x)$ or bandwidth $h(x)$ against $\rho(X)$, showing whether the scale estimator increases as the station approaches saturation.

6.7.4 Methods and metrics

The main ablation will compare two CKME-DCP variants: fixed bandwidth and plug-in scale-adaptive bandwidth with $h(x) = c\widehat{s}(x)$. If implementation time permits, DCP-DR or CQR can be included as external conformal baselines, but they should not distract from the main ablation: whether scale-adaptive CKME-DCP improves interval behavior in a realistic high-congestion ROI. The reported metrics will be target marginal coverage under q_X, utilization-bin coverage, mean interval width, Winkler interval score, and the profile of $\widehat{s}(x)$ or $h(x)$ against utilization.

The results should be reported in three blocks. The first block should show target marginal coverage and interval score, because these are the main conformal and efficiency summaries. The second block should show utilization-bin coverage, because it reveals whether a method is unstable within the ROI. The third block should show mechanism diagnostics: the empirical relationship between utilization, waiting-time spread, $\widehat{s}(x)$, and $h(x)$. The intended story is that scale adaptivity is not just changing one average number; it is aligning the smoothing scale with the congestion-driven output scale.

The expected manuscript role of this case study is diagnostic rather than theorem-extending. It should show that, in a realistic simulation setting where waiting-time variability changes sharply across low- and high-congestion regimes, scale-adaptive CKME-DCP can preserve target coverage while improving bin-wise stability and interval efficiency relative to a fixed-bandwidth CKME-DCP baseline. Numerical tables and figures will be inserted here after the EV charging simulation runs are completed.

7 Positioning Relative to Existing CP Methods

The proposed method should not be presented as "CKME makes conformal prediction valid" or as a method that automatically handles arbitrary target distributions. Conformal validity is already estimator-agnostic, and target validity requires matched calibration or a valid reweighting argument. The correct positioning is narrower and stronger:

target-aware calibration specifies where validity holds,
scale-adaptive CKME improves the DCP score used for efficiency.

Conformalized quantile regression adapts interval endpoints through estimated conditional quantiles [12]. Distribution regression versions of DCP estimate the conditional CDF through threshold-indexed regressions [5]. Localized and weighted conformal methods modify the calibration distribution or score weighting to improve local guarantees [16, 6]. CKME-DCP is complementary: it estimates a single conditional distributional object and uses smooth test functions to evaluate CDFs. This representation creates a bandwidth parameter whose local scaling can be matched to the response scale.

This distinction also clarifies the role of adaptive design. Adaptive site selection changes where simulation data are collected. It is a data-acquisition layer and can improve estimator quality in high-uncertainty regions, but it is not the same contribution as scale-adaptive smoothing. A future extension can combine the present target-aware calibration framework with design-aware acquisition rules, but the current paper keeps the main theorem line focused on target validity and scale-adaptive scoring.

8 Discussion and Next Steps

The current draft converts the WSC paper into a broader journal story by moving the target distribution and scale bandwidth issues to the front of the paper. The WSC version established the core CKME-DCP pipeline and the basic split-conformal guarantee. The journal version should now emphasize three points.

First, the coverage statement is only meaningful after the target input distribution is specified. Global-domain coverage, region-of-interest coverage, fixed-target-set average coverage, and smoothed fixed-target local coverage are different claims. The target joint law $P^q_{XY} = q_X P_{Y|X}$ should therefore be stated before the algorithm rather than added as a limitation at the end.

Second, the adaptive bandwidth is not merely an empirical tuning trick. Under heteroscedasticity, fixed smoothing creates the standardized ratio $h/s(x)$. Setting $h(x) = c\widehat{s}(x)$ is a principled way to align CDF resolution across the input space.

Third, future experiments should be organized around calibrated interval performance rather than raw CDF estimation alone. Raw CDF diagnostics can explain mechanism, but the paper's inferential target is coverage, width, interval score, and group-conditional stability after conformal calibration. If q_X is promoted from a formulation layer to a full contribution layer, the next empirical step should include an ROI or fixed-target experiment in addition to the current scale-adaptive bandwidth experiments.

A Proofs

A.1 Proof of Proposition 1

Condition on the training data and on the fitted estimator $\widehat{F}$. Under the stated exchangeability condition, the calibration scores

$$S_i^{\text{cal}} = S_{\widehat{F}}(X_i^{\text{cal}}, Y_i^{\text{cal}}), \qquad i = 1, \dots, n_{\text{cal}},$$

and the target score

$$S_\star = S_{\widehat{F}}(X_\star, Y_\star)$$

are exchangeable. Let

$$k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil .$$

If $k \le n_{\text{cal}}$, $\widehat{q}_{1-\alpha}$ is the k-th order statistic of the calibration scores; if $k = n_{\text{cal}} + 1$, the usual convention is $\widehat{q}_{1-\alpha} = +\infty$, in which case the claim is trivial. For $k \le n_{\text{cal}}$, the rank of $S_\star$ among the $n_{\text{cal}} + 1$ exchangeable scores is uniform up to ties, and the conservative tie convention gives

$$\mathbb{P}\{S_\star \le \widehat{q}_{1-\alpha} \mid \mathcal{D}_{\text{tr}}, \widehat{F}\} \ge \frac{k}{n_{\text{cal}} + 1} \ge 1 - \alpha.$$

Since $Y_\star \in \widehat{C}(X_\star)$ is exactly the event $S_\star \le \widehat{q}_{1-\alpha}$, taking expectation over the training data proves

$$\mathbb{P}_{(X,Y)\sim P_{XY}^q}\{Y \in \widehat{C}(X)\} \ge 1 - \alpha.$$

A.2 Proof of Proposition 2

Let $A(u) = |u - 1/2|$. This map is one-Lipschitz. On the event in Assumption 1,

$$\begin{aligned} \left|S_{\widehat{F}}(x, y) - S^\circ(x, y)\right| &= \left|A(\widehat{F}(y \mid x)) - A(F^\circ(y \mid x))\right| \\ &\le \left|\widehat{F}(y \mid x) - F^\circ(y \mid x)\right| \le \epsilon_n \end{aligned}$$

uniformly over $x \in \mathcal{X}$ and $y \in \mathcal{T}$. This proves the score perturbation bound.

For the threshold statement, define the oracle calibration scores

$$S_i^\circ = S^\circ(X_i^{\text{cal}}, Y_i^{\text{cal}}), \qquad i = 1, \dots, n_{\text{cal}},$$

and let $q_{n,1-\alpha}^\circ$ be their k-th order statistic, with the same $k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$. The uniform score perturbation bound implies

$$\max_{1 \le i \le n_{\text{cal}}} \left|\widehat{S}_i - S_i^\circ\right| \le \epsilon_n.$$

Order statistics are one-Lipschitz with respect to the sup-norm perturbation of the sample, hence

$$\left|\widehat{q}_{1-\alpha} - q_{n,1-\alpha}^\circ\right| \le \epsilon_n.$$

By Assumption 2, the oracle score distribution is regular at $q_{1-\alpha}^\circ$. Standard empirical-quantile theory gives

$$\left|q_{n,1-\alpha}^\circ - q_{1-\alpha}^\circ\right| = O_{\mathbb{P}}(n_{\text{cal}}^{-1/2}).$$

Combining the two displays yields

$$\left|\widehat{q}_{1-\alpha} - q_{1-\alpha}^\circ\right| = O_{\mathbb{P}}(\epsilon_n + n_{\text{cal}}^{-1/2}).$$

A.3 Proof of Theorem 1

Under $Y = m(x) + s(x)\varepsilon$ and $h(x) = cs(x)$,

$$F_c^\circ(y \mid x) = H_c\left(\frac{y - m(x)}{s(x)}\right).$$

Evaluating at $Y = m(x) + s(x)\varepsilon$ gives

$$F_c^\circ(Y \mid X = x) = H_c(\varepsilon).$$

Because $\varepsilon \perp X$, the conditional distribution of $H_c(\varepsilon)$ given $X = x$ is the same for all x. Applying any measurable transform A preserves equality in distribution, so

$$S_c^\circ(X, Y) \mid X = x = A(F_c^\circ(Y \mid X = x)) \overset{d}{=} A(H_c(\varepsilon)).$$

A.4 Proof of Proposition 3

Write

$$z_x(t) = \frac{t - m(x)}{s(x)}.$$

Under the location-scale model,

$$\frac{t - Y}{c\widehat{s}(x)} = \frac{z_x(t) - \varepsilon}{c} \frac{s(x)}{\widehat{s}(x)}, \qquad \frac{t - Y}{cs(x)} = \frac{z_x(t) - \varepsilon}{c}.$$

On the event $\Delta_n < 1/2$,

$$\sup_{x \in \mathcal{X}} \left| \frac{s(x)}{\widehat{s}(x)} - 1 \right| = \sup_{x \in \mathcal{X}} \left| \frac{1}{\widehat{s}(x)/s(x)} - 1 \right| \leq 2\Delta_n.$$

Using the Lipschitz property of Ψ, for every x, t,

$$\begin{aligned}
&\left| F_{\widehat{h}}^\circ(t \mid x) - F_c^\circ(t \mid x) \right| \\
&\quad \leq \mathbb{E}\left[L_\Psi \left| \frac{z_x(t) - \varepsilon}{c} \right| \left| \frac{s(x)}{\widehat{s}(x)} - 1 \right| \middle| X = x \right] \\
&\quad \leq \frac{2L_\Psi}{c} \Delta_n \mathbb{E}\left[|z_x(t) - \varepsilon| | X = x \right].
\end{aligned}$$

Taking the supremum over $x \in \mathcal{X}$ and $t \in \mathcal{T}$, and applying Assumption 4, gives

$$\sup_{x,t} \left| F_{\widehat{h}}^\circ(t \mid x) - F_c^\circ(t \mid x) \right| \leq \frac{2L_\Psi M_{\mathcal{T}}}{c} \Delta_n$$

with probability tending to one. Assumption 3 gives the result.

A.5 Proof of Proposition 4

For every x, t,

$$\left| \widehat{F}_{h(\cdot)}(t \mid x) - F_{h(\cdot)}^\circ(t \mid x) \right| = \left| \langle g_{t,h(x)}, \widehat{\mu}_{Y|x} - \mu_{Y|x} \rangle_{\mathcal{H}_Y} \right| \leq \left\| g_{t,h(x)} \right\|_{\mathcal{H}_Y} \left\| \widehat{\mu}_{Y|x} - \mu_{Y|x} \right\|_{\mathcal{H}_Y}.$$

Taking suprema and applying Assumptions 5–6 gives the result.

A.6 Proof of Theorem 2

Let

$$\widehat{S}(x,y) = S_{\widehat{F}}(x,y), \qquad S_c^\circ(x,y) = \left|F_c^\circ(y \mid x) - 1/2\right|.$$

By Proposition 4 and Proposition 2, there exists a sequence $r_n = o_{\mathbb{P}}(1)$ such that, with probability tending to one,

$$\sup_{x,y}\left|\widehat{S}(x,y) - S_c^\circ(x,y)\right| \le r_n, \qquad \left|\widehat{q}_{1-\alpha} - q_{1-\alpha}^\circ\right| \le r_n.$$

On this event, for any group $\mathcal{G}_k$,

$$\{S_c^\circ(X,Y) \le q_{1-\alpha}^\circ - 2r_n\} \subseteq \{\widehat{S}(X,Y) \le \widehat{q}_{1-\alpha}\} \subseteq \{S_c^\circ(X,Y) \le q_{1-\alpha}^\circ + 2r_n\}.$$

Therefore,

$$\begin{aligned}&\left|\mathbb{P}\{\widehat{S}(X,Y) \le \widehat{q}_{1-\alpha} \mid X \in \mathcal{G}_k\} - \mathbb{P}\{S_c^\circ(X,Y) \le q_{1-\alpha}^\circ \mid X \in \mathcal{G}_k\}\right| \\ &\quad \le \sup_{\left|u - q_{1-\alpha}^\circ\right| \le 2r_n}\left|G_k(u) - G_k(q_{1-\alpha}^\circ)\right|,\end{aligned}$$

where G_k is the groupwise oracle score CDF in Assumption 7. By the uniform Lipschitz condition, the right-hand side is $O(r_n) = o_{\mathbb{P}}(1)$, uniformly over the fixed finite set of groups.

It remains to identify the oracle group coverage level. Theorem 1 implies that the conditional distribution of $S_c^\circ(X,Y)$ given $X = x$ does not depend on x. Hence every group has the same oracle score distribution as the marginal target distribution:

$$\mathbb{P}\{S_c^\circ(X,Y) \le u \mid X \in \mathcal{G}_k\} = \mathbb{P}\{S_c^\circ(X,Y) \le u\}.$$

At the continuity point $q_{1-\alpha}^\circ$, this probability equals $1-\alpha$. Thus

$$\max_{1 \le k \le K}\left|\mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{G}_k\} - (1-\alpha)\right| \to 0$$

in probability.

A.7 Proof of Corollary 1

Let $F_{\widehat{h}}^\circ$ denote the plug-in smoothed oracle CDF and let F_c° denote the oracle adaptive smoothed CDF. By the assumed uniform CKME consistency for $F_{\widehat{h}}^\circ$,

$$\sup_{x,t}\left|\widehat{F}_{h(\cdot)}(t \mid x) - F_{\widehat{h}}^\circ(t \mid x)\right| = o_{\mathbb{P}}(1).$$

By Proposition 3,

$$\sup_{x,t}\left|F_{\widehat{h}}^\circ(t \mid x) - F_c^\circ(t \mid x)\right| = o_{\mathbb{P}}(1).$$

The triangle inequality gives

$$\sup_{x,t}\left|\widehat{F}_{h(\cdot)}(t \mid x) - F_c^\circ(t \mid x)\right| = o_{\mathbb{P}}(1).$$

Thus Assumption 1 holds with $F^\circ = F_c^\circ$, up to replacing ϵ_n by the sum of the CKME and plug-in scale errors. The proof of Theorem 2 then applies verbatim.

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